

Cloud Computing Systems

Dawid Maciejewski,
Yevhen Martyn,
Maksym Huryn,
Michał Wawrzko





Problem Statement

Design a data exchange system between an application and the cloud system. The system should have an ability to transfer and store files, implement file version control and manage login, authentication and authorization of users.



Objectives

The system should be able to:

- Login/Logout
 - Download file
 - Upload file
 - Read list of files
 - Send many files as one packed file
 - Logbook and registration of logins and operations
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Research

Our team used the following sources:

- Web search engines (Google, DuckDuckGo)
 - AI models for quick aggregation of solutions (OpenAI)
 - Third-party websites (alternativeto.net, libhunt.com)
 - Review with the experts in person
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Existing Solutions

	Azure 	AWS 	Self-made file server
Licensing	Proprietary; subscription.	Proprietary; subscription.	Open-source, no licensing fees.
Scalability	Highly scalable; automated	Highly scalable; automated	Manual scaling, hardware-limited.
Cost	Pay-as-you-go	Pay-as-you-go	Free
Customization	Limited to Azure tools.	Limited to AWS tools.	Fully customizable
Use Cases	Enterprise, hybrid cloud.	Broad, scalable workloads.	Small-scale, privacy-focused setups.

- Seafile is an open-source cloud storage system designed for file synchronization, sharing, and collaboration.
- It combines the capabilities of a private cloud with powerful collaboration tools, making it suitable for both individuals and organizations.

Why We Chose Seafile

Unlike public cloud services, Seafile on a VM provides:

- **Greater control over data.**
- **Cost efficiency** by avoiding recurring subscription fees.
- **Customization** to fit our specific requirements.



Seafile



Toolchain

- Virtual Machine Monitor: VirtualBox
 - OS: Ubuntu Server 22.04.5
 - Database: MySQL 8.0.39-0
 - File Hosting Service: Seafile 11.0.12
 - HTTPS Server: Apache2
 - SSH Server: OpenSSH
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Challenges

- Choosing and setting up a Linux server distro
 - Setting up all services (Apache2, MySQL Server etc.)
 - Synchronization of the workflow
 - Encryption (HTTPS)
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Risk and Measures of Success

Risks:

- Overwriting somebodies progress
- Failure to implement one (or more) of the requirements (especially encryption and uploading multiple files as ZIP)
- System downtime (server crashes, high traffic, blackout)
- Scalability issues due to hardware limitations

Measures of Success:

- Image of the virtual machine containing its functions that's easily set up.
 - Fast response times for file transfers
 - Secure connection (encrypted)
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Thank you for
attention!